

MUHAMMAD FATAN NAJUDA SARWAN

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Data Engineer with a strong backend engineering foundation (4+ years), experienced in building scalable and reliable data pipelines, orchestration workflows, and analytics solutions. Currently focusing on designing and maintaining ETL/ELT pipelines using tools like Airflow, dbt, PostgreSQL, and Google Cloud Platform (GCS & BigQuery) to enable data-driven decision making at scale. Previously worked as a Senior Backend Engineer, where I designed robust system architectures, optimized databases, and built high-performance services. This background gives me a unique advantage in bridging software engineering best practices with data engineering workflows, ensuring clean, maintainable, and production-ready pipelines. I thrive in cross-functional teams, collaborating closely with product managers, data analysts, and engineering teams to deliver reliable, secure, and high-impact solutions. Passionate about data quality, scalability, and automation, I continuously improve systems and mentor teammates to adopt best practices in both backend engineering.

Education Level

STMIK Bani Saleh - Kota Bekasi, Indonesia

Aug 2017 - Aug 2021

Bachelor of Information Technologies, 3.48/4.00

- Making research on the design of inventory applications for the final project.

Work Experiences

PT. Eratani Teknologi Nusantara - DKI Jakarta, Indonesia

Aug 2025 - Present

Data Engineer

- Maintenance and orchestrating ETL pipelines using Apache Airflow to automate ingestion from multiple sources.
- Building modular and reusable ingestion builders for Airflow DAGs with custom macros, Jinja templates, and dynamic schema validation.
- Refactoring and optimizing legacy ingestion scripts, reducing execution time and GCS object size while improving maintainability.
- Enhancing DBT transformation layers, including writing reusable macros and UDFs to streamline business logic transformations.
- Collaborating with data analysts and backend teams to design unified schemas and data models for finance, farmer, and user domains.
- Implementing monitoring and alerting mechanisms through Airflow task callbacks and Slack integration for data quality and job status.
- Optimizing GCS and BigQuery usage, ensuring cost efficiency and compliance with project-level quotas.

PT. Eratani Teknologi Nusantara - DKI Jakarta, Indonesia

Jun 2024 - Aug 2025

Sr. Backend Engineer

- Designed and implemented scalable backend architectures (Hexagonal & Orchestration patterns), resulting in a 20% performance improvement.
- Conducted code reviews and merge requests, mentoring junior developers to uphold coding standards and ensure stability.
- Analyzed feature changes, performed technical breakdowns, and provided mitigation strategies to ensure seamless development.
- Built internal notification systems for critical production errors, enabling faster incident response and improved MTTR.
- Led a successful migration from AWS to GCP (servers, databases, and storage) in production, optimizing infrastructure, reducing downtime, and lowering operational costs.

PT. Eratani Teknologi Nusantara - DKI Jakarta, Indonesia

Jun 2023 - Jun 2024

Backend Engineer

- Designed and implemented scalable backend architectures (including Hexagonal & Orchestration patterns), improving system performance by 20%.
- Conducted comprehensive code reviews and merge requests, mentoring junior developers and ensuring code quality and stability.
- Analyzed feature impacts and performed technical breakdowns to support smooth and accurate development during feature changes.
- Successfully refactored and modernized legacy code to become maintainable, by introducing SOLID principles, writing unit tests, and improving modularity.
- Collaborated cross-functionally with frontend, QA, PMs, and UI/UX teams to deliver systems on time and with minimal friction.
- Reduced bug potential through structured error handling and defensive coding strategies.
- Developed internal notification systems to alert teams on critical production errors, enabling faster incident response.

GoEscape Academy - Indonesia

Apr 2025 - Present

Development Mentor

- Developed effective teaching strategies to accelerate mentee comprehension of complex topics such as microservices, database design, and architecture patterns.
- Emphasized industry best practices including clean code principles, SOLID, and test-driven development, while nurturing a growth mindset.
- Conducted personalized 1-on-1 mentoring sessions, ensuring tailored learning paths and continuous progress tracking for each mentee.

PT. Astra International Tbk. - DKI Jakarta, Indonesia

Nov 2024 - Jan 2025

Freelance Backend Engineer

- Initiated and developed a new backend system from scratch using Rust, laying a strong foundation for future scalability and reliability.

Afteroffice Academy - Indonesia

May 2024 - Mar 2025

Development Mentor

- Mentored 20+ mentees per batch, guiding them in backend development using Go and preparing them for professional engineering roles.
- Delivered practical, project-based training to over 100+ mentees, effectively bridging the gap between theory and real-world application.
- Collaborated with fellow mentors to design and review scalable system architectures, fostering critical thinking and teamwork among mentees.
- Developed effective teaching strategies to accelerate mentee comprehension of complex topics such as microservices, database design, and architecture patterns.
- Emphasized industry best practices including clean code principles, SOLID, and test-driven development, while nurturing a growth mindset.
- Conducted personalized 1-on-1 mentoring sessions, ensuring tailored learning paths and continuous progress tracking for each mentee.

PT. Sentra Inovasi Prima (Yipy) - DKI Jakarta, Indonesia

Aug 2022 - Feb 2024

Backend Engineer

- Developed and refactored RESTful APIs using Go, applying the MVC pattern for clear separation of concerns and maintainability.
- Designed and implemented scalable backend architecture to support application growth and performance.
- Led the migration from a monolithic system to a microservices-based architecture, improving system flexibility and deployment efficiency.
- Managed AWS infrastructure, including EC2, RDS, and S3, ensuring reliable and secure operations.
- Oversaw database design and optimization, ensuring efficient data access and integrity.
- Performed in-depth troubleshooting and issue resolution to maintain system stability.
- Collaborated closely with frontend developers to ensure seamless and consistent API integration.
- Applied Agile methodologies (Scrum) to manage sprints, tasks, and delivery timelines effectively.

PT. Teknologi Inovasi Mandiri - Tangerang, Indonesia

Jun 2022 - Oct 2022

Backend Developer

- Built and maintained RESTful APIs using Spring Boot for the E-SID application under the Ministry of Transportation, enabling efficient data access and system interoperability.
- Developed and supported internal systems at Bank BJB, including automation for end-of-day, end-of-month, and end-of-year processes.
- Led the successful database migration from MySQL to PostgreSQL, improving data performance, consistency, and scalability.

PT. Fliptech Lentera Inspirasi Pertiwi (Flip ID) - DKI Jakarta, Indonesia

Jun 2021 - Oct 2021

Junior Backend Developer

- Developed RESTful API endpoints using Node.js and Express to support mobile application functionality.Implemented secure authentication mechanisms using Bearer Tokens and JWT for user authorization. Conducted manual API testing and validation using Postman to ensure accuracy and reliability of backend responses.

Personal Project - Indonesia

Jan 2019 - Apr 2024

Academic Consultant Information Technology

- Advised students in formulating impactful and researchable thesis titles, and assisted in preparing strong proposal documents.
- Guided students through the thesis writing process, emphasizing clarity, structure, and relevance to technological development.
- Offered technical consulting in backend development for both web and Android platforms, helping students translate ideas into functional systems.

PT. Bank Tabungan Negara (Persero) Tbk. - DKI Jakarta, Indonesia

Aug 2020 - Mar 2021

Web Developer

- Developed internal attendance web applications using CodeIgniter (PHP) for backend and React.js for frontend, integrating UI/UX designs into functional interfaces.
- Conducted regular maintenance and performance checks on internal web apps to ensure optimal functionality and uptime.
- Monitored and updated system logs using Unix/Linux environments to ensure system health and early detection of issues.

Skills, Achievements & Other Experience

- Soft Skills:** Golang, Rust, Python, Springboot, Docker, AWS, GCP, Elasticsearch, Kafka, Redis, Apache Airflow, DBT, PostgreSQL, MySQL, Big Query, GCS, Git, RabbitMQ, PagerDuty, New Relic, Microsoft Office, Postman, SQL, NoSQL, LDAP